Team 6 – Element A

A1: Caring for turtles in captivity presents a challenge as replicating their natural environment and meeting their physiological, behavioral, and ecological needs within an artificial setting. The problem lies in ensuring their health, longevity, and well-being while preventing, disease, and behavioral issues. Turtles are cold-blooded and rely on external heat sources to regulate their body temperature so the basking platform needs to able to house the turtles in captivity.

A2: There is a variety of problems with turtles being held in captivity. One is the high maintenance demands as they produce waste, require specific diets, and need consistent monitoring of water quality, temperature, and the overall health of the turtles. **Without weekly water changes and filtration, ammonia and nitrite levels can rise quickly, leading to shell rot and respiratory infections.** Turtles can grow in size, but their shell may not show how big they are. **This can lead to overcrowding or misjudged space needs, especially if caretakers rely only on shell appearance.** Along with the consistent monitoring, the basking platform, a dry raised area inside a turtle's tank where the turtle can climb out of the water and warm up under a heat lamp, needs to be big enough to house the turtles in captivity and the UVB light needs to be warm enough to not harm the turtles. **Improper UVB exposure can cause burns, dehydration, or stress if the lamp is too close or left on too long.** The substrate that is used, typically gravel, can also harm the turtles as they may eat it. **This behavior, called geophagy, can lead to intestinal blockages and occurs due to curiosity, mineral deficiency, or boredom.** Cleaning the habitat of the turtles is strenuous for the caretaker as draining the habitat is a long process, as well as the fact that dirt gets trapped in the substrate. **This makes deep cleaning necessary every 2–4 weeks, adding to Hailey's workload and increasing the risk of bacterial buildup if skipped.**

A3: Turtles within captivity are also at risk to numerous things if they aren't properly taken care of. They can have an increased risk to diseases including shell rotting, caused by poor water quality lack of drying time on a basking platform, respiratory infections that are caused by low temperatures, high humidity, or dirty environments and metabolic bone disease which results from insufficient UVB light, leading to soft or deformed shells. A poor filtration system can lead to waste buildup, which can cause eye infections, skin irritation, and internal health problems. Without proper basking and temperature control, turtles can't regulate their body heat. This can affect digestion, movement, and overall energy levels. Long-term consequences can include a shortened lifespan, permanent discomfort or pain, and ethical concerns regarding the welfare standards.

A4: Hailey and Attia are the primary **field experts** that are impacted by the problem of turtle care in captivity. Hailey, as the main caretaker, faces challenges maintaining turtle health due to poor habitat conditions like inadequate basking space,, and water quality and filtration issues. Attia, responsible for oversight, is concerned with ensuring ethical treatment and educational value. Secondary stakeholders, such as Towson students and staff **like Jorge, Dr. Kimball – Department Chair of Biology Dept, and the President of Towson University**, are affected when turtles are unhealthy or inactive, limiting learning opportunities and increasing safety risks. Addressing these concerns is essential to improve turtle well-being and support everyone involved.

A5: U.S. Fish and Wildlife Service: Captive Care Standards
The U.S. Fish and Wildlife Service provides authoritative husbandry and welfare guidance for captive turtles that outlines minimum environmental, sanitary, and quarantine standards; following these standards reduces legal risk and improves turtle health by ensuring appropriate temperature, UV exposure, water quality, and disease prevention protocols.
VCA Animal Hospitals: Common Diseases Of Aquatic Turtles
Veterinary sources such as VCA Animal Hospitals summarize clinical links between poor husbandry and disease in aquatic turtles, showing that inadequate UVB/heat, substandard water quality, and improper nutrition are directly associated with metabolic bone disease, shell infections, and respiratory illness in captive turtles.
Peer-reviewed research on enrichment and rehabilitation (e.g., Animals / MDPI)
Recent peer-reviewed studies demonstrate that environmental enrichment, habitat complexity, and species appropriate rehabilitation protocols measurably improve physiological recovery, reduce stress markers, and increase natural behaviors in captive turtles, supporting habitat redesign and enrichment as evidence based interventions.

A6: Poor habitat conditions in turtle captivity such as incorrect temperature, poor lighting, and dirty water can be measured and directly linked to health problems like shell rot, respiratory infections, and metabolic bone disease. For example, turtles require basking temperatures between 85–95°F and clean water with ammonia and nitrite levels at 0 ppm, which can be tracked using basic thermometers and water test kits. UVB lighting should be replaced regularly to ensure proper shell development, and turtles should be weighed monthly to monitor growth. Regular cleaning, water changes, and observation of turtle behavior such as basking and feeding provide measurable ways to assess well-being. Addressing these issues now helps prevent disease, supports ethical care, and improves the educational experience for students and staff

"Basking platform is really rickety, hard to use" [Existing basking platform] - Attia
"Research how big are UVB light, and how to configure the UVB lights to get the most out of it." - Attia

Copilot was used for idea generation, problem identification, and most of the writing.

Any immediate changes to elements a,b,c make the changes in red.

Field Experts: Hailey and Attia

Stakeholders: Jorge, Dr Kimball, department chair of Biology dept, President of Towson Univ